

Experiment 11.1: Potential Difference, Current and Resistance

Notes

Background

The simplest way of thinking about an operating electric circuit is to consider the potential difference to be the *cause* and the current to be the *effect*. The resistance is the property of a circuit component that determines the amount of current that will flow for a given potential difference. The resistance of a circuit component determines the way it will behave in a particular location in that circuit. The energy dissipated by a component depends on its resistance and the potential difference it experiences when placed into a circuit.

A simple example of how this might be used in designing a device is the selection of a heating element for a kettle, a heater, or a stove. In each case, the potential difference will be 240 V and a particular output of heat will be required.

Aim

To investigate the relationship between the applied difference and the current in a resistor.

An element of the correct resistance will need to be selected in order that the correct power output is obtained.

Apparatus (per group)

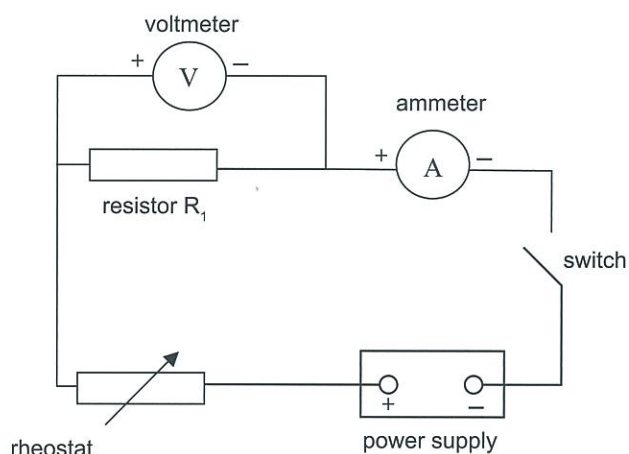
- power supply
- switch
- seven electrical leads
- voltmeter or multimeter
- ammeter or multimeter
- three different resistors (you can also use diodes and LEDs)
- variable resistor (rheostat)

Pre-Lab

- In this practical exercise you will use electrical meters to make measurements of potential difference and current in order to calculate the resistance of some resistors.
- The voltmeter measures the potential difference across a component. This means that the *voltmeter must be connected in parallel* with that component.
- The ammeter measures the current through a component. This means that the *ammeter must be connected in series* with that component.
- Most multimeters can measure DC or AC potential difference, DC or AC current, or resistance depending on the measurement mode selected. You must still take care to connect the meter correctly when measuring potential difference and current.
- In your group, discuss how you will calculate the resistance of each resistor from the measurements you have made. If you use a multimeter, you will be able to check your resistance calculations by direct measurement.

Lab Notes

- Connect the equipment as shown in the circuit diagram (*below*). Make sure that the meters are connected with the correct polarity. Your teacher may wish to check this before you close the switch.
- Set the power supply to the lowest voltage setting.
- Adjust the power supply and rheostat to get a number of different readings of current and voltage (potential difference).
- Record your measurements in a suitable table.
- Note: Do not run the circuit for more than ten seconds at a time. The current will cause an increase in the temperature of the resistor, giving unexpected results.
- Repeat steps 2 and 3 with your other two resistors.



Apparatus for Experiment 16.1

Post-Lab Discussion

1. Calculate and record the resistance values, using:
$$R = \frac{V}{I}$$
 for each set of readings.
2. Plot a graph of current against potential difference for each resistor.
3. When the current is directly proportional to the potential difference, a conductor is called an 'ohmic conductor'. Were any of your resistors ohmic?
4. How do the graphs for each resistor differ? Why are they different?
5. What types of materials are ohmic?

Notes